

# SSH - Secure Shell

Registration No. : 2025/CS/079

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## Introduction to SSH

### Activity 01

( 01 )

```
[chamal@ubuntu:~$ sudo apt install openssh-server
openssh-server is already the newest version (1:10.2p1-2ubuntu3.2).
openssh-server set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 16
chamal@ubuntu:~$
```

( 02 )

```
[chamal@ubuntu:~$ sudo systemctl start ssh
[chamal@ubuntu:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
chamal@ubuntu:~$
```

( 03 )

```
[chamal@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 2a:a1:4c:0f:78:89 brd ff:ff:ff:ff:ff:ff
    altname enx2aa14c0f7889
    inet 192.168.64.24/24 metric 100 brd 192.168.64.255 scope global dynamic enp0s1
        valid_lft 2801sec preferred_lft 2801sec
    inet6 fd19:5974:35dd:731b:28a1:4cff:fe0f:7889/64 scope global dynamic mngtmpaddr no
    prefixroute
        valid_lft 2591994sec preferred_lft 604794sec
    inet6 fe80::28a1:4cff:fe0f:7889/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
chamal@ubuntu:~$
```

( 04 )

```
Mac Terminal
[chamal@MacBook-Air: ~ $ ssh chamal@192.168.64.24
[chamal@192.168.64.24's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-22-generic aarch64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Jun 26 01:45:54 PM +0530 2026

System load:          0.01
Usage of /:           22.8% of 47.47GB
Memory usage:         48%
Swap usage:           0%
Processes:            227
Users logged in:      0
IPv4 address for enp0s1: 192.168.64.24
IPv6 address for enp0s1: fd19:5974:35dd:731b:28a1:4cff:fe0f:7889

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

Expanded Security Maintenance for Applications is not enabled.
```

## Activity 02

( 01 )

```
[chamal@ubuntu:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/chamal/.ssh/id_ed25519):
Enter passphrase for "/home/chamal/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/chamal/.ssh/id_ed25519
Your public key has been saved in /home/chamal/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:EdDqZm0cQQIetazo8dIo+VzIqX55Ya0ea5hvoj9zcf0 chamal@ubuntu
The key's randomart image is:
+--[ED25519 256]--+
|  oo. .o.         |
| . o... ..        |
| . oo ..          |
| . . o .          |
|... o +S          |
|.o=o.+X .         |
|o+*o=*o. .        |
|.B+Bo.   E        |
|=o@Oo             |
+----[SHA256]-----+
chamal@ubuntu:~$
```

( 02 )

```
chamal@ubuntu:~$ ssh-copy-id chamalinduware@10.23.59.71
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host '10.23.59.71 (10.23.59.71)' can't be established.
ED25519 key fingerprint is: SHA256:dr9CSmiqIJDZno0lDihVQQqIjbvkb4iMf+W4xhNvEjs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
The authenticity of host '10.23.59.71 (10.23.59.71)' can't be established.
ED25519 key fingerprint is: SHA256:dr9CSmiqIJDZno0lDihVQQqIjbvkb4iMf+W4xhNvEjs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install all the new keys
(chamalinduware@10.23.59.71) Password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'chamalinduware@10.23.59.71'"
and check to make sure that only the key(s) you wanted were added.
```

( 03 )

```
chamal@ubuntu:~$ ssh chamalinduware@10.23.59.71
chamal@MacBook-Air: ~ $
```

### Activity 03

( 01 )

```
[ chamal@ubuntu:~$ echo "Hello SSH" > hello.txt
chamal@ubuntu:~$
```

( 02 )

```
[ 1 chamal@ubuntu:~$ scp hello.txt chamalinduware@10.23.59.71:/Users/chamalinduware
hello.txt                               100% 10 10.9KB/s 00:00
chamal@ubuntu:~$ exit
logout
Connection to 192.168.64.24 closed.
chamal@MacBook-Air: ~ $ ls | grep hello
hello.txt
chamal@MacBook-Air: ~ $ cat hello.txt
Hello SSH
chamal@MacBook-Air: ~ $
```

( 03 )

```
chamal@ubuntu:~$ scp chamalinduwara@10.23.59.71:/Users/chamalinduwara/hello.txt /home/c  
hamal/Desktop/  
hello.txt 100% 10 4.4KB/s 00:00  
chamal@ubuntu:~$ ls Desktop/ | grep hello  
hello.txt  
chamal@ubuntu:~$
```

#### Activity 04

( 01 )

```
chamal@MacBook-Air: ~ $ ssh chamal@192.168.64.24 "df -h"  
Enter passphrase for key '/Users/chamalinduwara/.ssh/id_ed25519':  
chamal@192.168.64.24's password:  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           677M  2.2M  674M   1% /run  
/dev/mapper/ubun--vg-ubuntu--lv 48G   11G   35G  24% /  
tmpfs           1.7G   40K   1.7G   1% /dev/shm  
efivarfs        256K   30K   227K  12% /sys/firmware/efi/efivars  
none            1.0M    0    1.0M   0% /run/credentials/systemd-journ  
ald.service  
tmpfs           1.7G   8.0K   1.7G   1% /tmp  
none            1.0M    0    1.0M   0% /run/credentials/systemd-resol  
ved.service  
/dev/vda2       2.0G  120M   1.7G   7% /boot  
/dev/vda1       1.1G   6.6M   1.1G   1% /boot/efi  
none            1.0M    0    1.0M   0% /run/credentials/systemd-netwo  
rkd.service  
none            1.0M    0    1.0M   0% /run/credentials/serial-getty@  
ttyAMA0.service  
tmpfs           339M   48K   339M   1% /run/user/1000  
chamal@MacBook-Air: ~ $
```

( 02 )

```
chamal@MacBook-Air: ~ $ ssh chamal@192.168.64.24 "cat /etc/os-release"  
Enter passphrase for key '/Users/chamalinduwara/.ssh/id_ed25519':  
chamal@192.168.64.24's password:  
PRETTY_NAME="Ubuntu 26.04 LTS"  
NAME="Ubuntu"  
VERSION_ID="26.04"  
VERSION="26.04 LTS (Resolute Raccoon)"  
VERSION_CODENAME=resolute  
ID=ubuntu  
ID_LIKE=debian  
HOME_URL="https://www.ubuntu.com/"  
SUPPORT_URL="https://help.ubuntu.com/"  
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"  
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"  
UBUNTU_CODENAME=resolute  
LOGO=ubuntu-logo  
chamal@MacBook-Air: ~ $
```



## SSH into virtual machine

### Activity 01

```
[chamal@ubuntu:~$ mkdir -p ~/project/reports/2025
[chamal@ubuntu:~$ tree project
project
├── reports
│   └── 2025

3 directories, 0 files
[chamal@ubuntu:~$ cd project
[chamal@ubuntu:~/project$ cd reports/
[chamal@ubuntu:~/project/reports$ cd 2025/
[chamal@ubuntu:~/project/reports/2025$ cd ~
[chamal@ubuntu:~$ pushd project
~/project ~
[chamal@ubuntu:~/project$ pushd reports/
~/project/reports ~/project ~
[chamal@ubuntu:~/project/reports$ pushd 2025/
~/project/reports/2025 ~/project/reports ~/project ~
[chamal@ubuntu:~/project/reports/2025$ popd
~/project/reports ~/project ~
[chamal@ubuntu:~/project/reports$ popd
~/project ~
[chamal@ubuntu:~/project$ popd
~
[chamal@ubuntu:~$ rmdir ~/project/reports/
rmdir: failed to remove '/home/chamal/project/reports/': Directory not empty
1 chamal@ubuntu:~$ rm -rf ~/project/reports/
chamal@ubuntu:~$
```

Rmdir command fails to remove the folder since it is not an empty directory (there is a directory called 2025 inside it)

But with `rm -rf` command it recursively delete everything inside the directory reports and then delete it.

## Activity 02

```
chamal@ubuntu:~$ touch log.txt data.csv config.ini
chamal@ubuntu:~$ echo "hello1" > log.txt
chamal@ubuntu:~$ echo "hello2" > data.csv
chamal@ubuntu:~$ echo "hello3" > config.ini
chamal@ubuntu:~$ ls -l
total 120
drwxr-xr-x 6 chamal chamal 4096 Jun 26 14:15 Desktop
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Documents
drwxr-xr-x 2 chamal chamal 4096 May 31 18:12 Downloads
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Music
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Pictures
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Public
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Templates
drwxr-xr-x 2 chamal chamal 4096 May 31 17:38 Videos
-rw-rw-r-- 1 chamal chamal 0 Jun 25 21:45 a.sh
-rw-rw-r-- 1 chamal chamal 4 Jun 25 21:23 a.txt
-rw-rw-r-- 1 chamal chamal 0 Jun 25 21:45 abcd.sh
drwxrwxr-x 4 chamal chamal 4096 Jun 10 21:24 assignment
-rw-rw-r-- 1 chamal chamal 0 Jun 25 21:45 b.py
-rw-rw-r-- 1 chamal chamal 4 Jun 25 21:23 b.txt
-rw-rw-r-- 1 chamal chamal 0 Jun 25 21:45 c.js
-rw-rw-r-- 1 chamal chamal 10 Jun 26 14:11 chamalinduwara@10.23.59.71
-rw-rw-r-- 1 chamal chamal 8 Jun 25 21:24 combined_output.txt
-rw-rw-r-- 1 chamal chamal 7 Jun 26 15:16 config.ini
drwxrwxr-x 2 chamal chamal 4096 Jun 25 21:20 csv_backup
-rw-rw-r-- 1 chamal chamal 7 Jun 26 15:15 data.csv
drwxrwxr-x 2 chamal chamal 4096 Jun 25 21:44 delete_test
drwxrwxr-x 3 chamal chamal 4096 Jun 12 15:53 demo_lab
-rwx----- 1 chamal chamal 0 Jun 25 21:16 final.lock
-rw-rw-r-- 1 chamal chamal 10 Jun 26 14:10 hello.txt
-rw-rw-r-- 1 chamal chamal 6 Jun 25 21:46 line1.txt
-rw-rw-r-- 1 chamal chamal 7 Jun 25 21:46 line2.txt
-rw-rw-r-- 1 chamal chamal 6 Jun 25 21:46 line3.txt
-rw-rw-r-- 1 chamal chamal 7 Jun 26 15:15 log.txt
-rw-rw-r-- 1 chamal chamal 0 Jun 8 08:28 main.c
drwxrwxr-x 2 chamal chamal 4096 Jun 12 15:25 newdir
drwxrwxr-x 2 chamal chamal 4096 Jun 26 15:12 project
drwxrwxr-x 2 chamal chamal 4096 Jun 25 21:20 project_space
drwxrwxr-x 3 chamal chamal 4096 Jun 25 21:28 sandbox
drwxrwxr-x 2 chamal chamal 4096 Jun 25 21:33 scripts
drwx----- 3 chamal chamal 4096 May 31 17:58 snap
```

```
chamal@ubuntu:~$ chmod 600 config.ini
chamal@ubuntu:~$ chmod 755 log.txt
```